| | Report | Result | Size | Time | Cycles | GPU | SM Frequency | Process | Attributes |
|---|---|---|---|---|---|---|---|---|---|
| Current | async | 573 - convolution3D_kernel | (24, 24, 1)x(32, 32, 1) | 45.15 us | 73,115 | 0 - NVIDIA GeForce RTX 4060 Laptop GPU | 1.62 GHz | [90991] conv3d_async | |
| Baseline 1 | baseline | 572 - convolution3D_kernel | (24, 24, 1)x(32, 32, 1) | 45.06 us | 72,913 | 0 - NVIDIA GeForce RTX 4060 Laptop GPU | 1.62 GHz | [104238] conv3d_baseline | |

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

## GPU Speed Of Light Throughput

GPU Throughput Chart

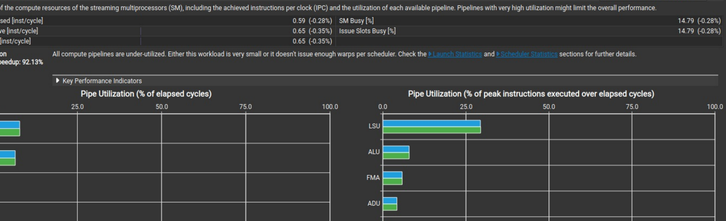
High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

| | | | |
|---|---|---|---|
| Compute (SM) Throughput [%] | 29.35 (-0.28%) | Duration [us] | 45.15 (+0.21%) |
| Memory Throughput [%] | 61.35 (-0.36%) | Elapsed Cycles [cycle] | 73,115 (+0.28%) |
| L1/TEX Cache Throughput [%] | 32.35 (-0.35%) | SM Active Cycles [cycle] | 64,317.38 (+0.35%) |
| L2 Cache Throughput [%] | 18.95 (-0.23%) | SM Frequency [Ghz] | 1.62 (+0.06%) |
| DRAM Throughput [%] | 61.35 (-0.36%) | DRAM Frequency [Ghz] | 7.99 (+0.14%) |

**High Memory Throughput** Memory is more heavily utilized than Compute: Look at the Memory Workload Analysis section to identify the DRAM bottleneck. Check memory replay (coalescing) metrics to make sure you're efficiently utilizing the bytes transferred. Also consider whether it is possible to do more work per memory access (kernel fusion) or whether there are values you can (re)compute.

Key Performance Indicators

**Roofline Analysis** The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 4% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.




## Compute Workload Analysis

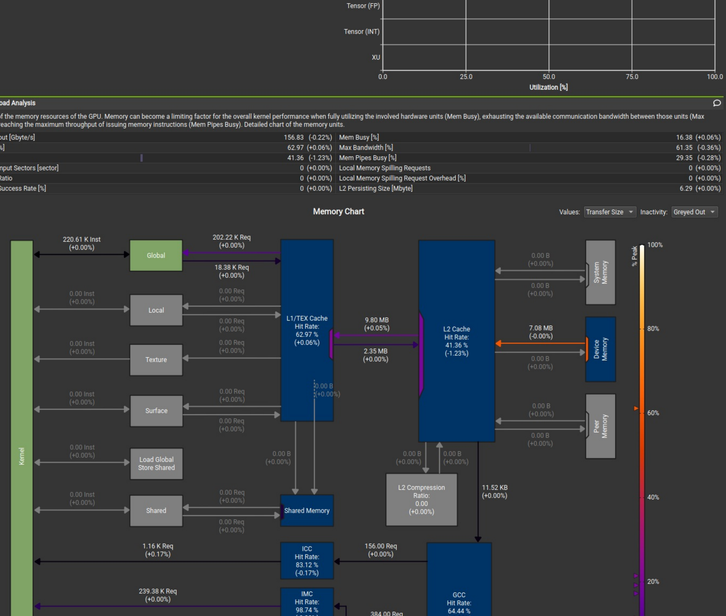
Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

| | | | |
|---|---|---|---|
| Executed Ipc Elapsed [inst/cycle] | 0.59 (-0.28%) | SM Busy [%] | 14.79 (-0.28%) |
| Executed Ipc Active [inst/cycle] | 0.65 (-0.35%) | Issue Slots Busy [%] | 14.79 (-0.28%) |
| Issued Ipc Active [inst/cycle] | 0.65 (-0.35%) | | |

**Low Utilization** All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.
**Est. Local Speedup: 92.13%**

Key Performance Indicators



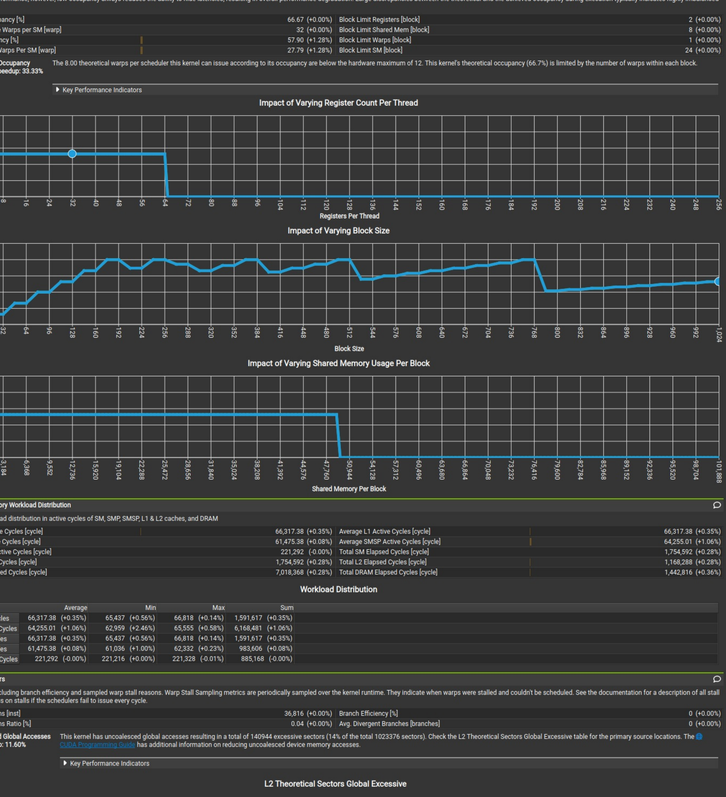

## Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units.

| | | | |
|---|---|---|---|
| Memory Throughput [Gbyte/s] | 156.83 (-0.22%) | Mem Busy [%] | 16.38 (+0.06%) |
| L1/TEX Hit Rate [%] | 62.97 (+0.06%) | Max Bandwidth [%] | 61.35 (-0.36%) |
| L2 Hit Rate [%] | 41.36 (-1.23%) | Mem Pipes Busy [%] | 29.35 (-0.28%) |
| L2 Compression Input Sectors [sector] | 0 (+0.00%) | Local Memory Spilling Requests | 0 (+0.00%) |
| L2 Compression Ratio | 0 (+0.00%) | Local Memory Spilling Request Overhead [%] | 0 (+0.00%) |
| L2 Compression Success Rate [%] | 0 (+0.00%) | L2 Persisting Size [Mbyte] | 6.29 (+0.00%) |

### Memory Chart

Values: Transfer Size | Inactivity: Greyed Out




## Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

| | | | |
|---|---|---|---|
| Grid Size | 576 (+0.00%) | Function Cache Configuration | CachePreferNone (CachePreferNone) |
| Registers Per Thread [register/thread] | 32 (+0.00%) | Static Shared Memory Per Block [byte/block] | 0 (+0.00%) |
| Block Size | 1,024 (+0.00%) | Dynamic Shared Memory Per Block [byte/block] | 0 (+0.00%) |
| Threads [thread] | 589,824 (+0.00%) | Driver Shared Memory Per Block [kbyte/block] | 1.02 (+0.00%) |
| Waves Per SM | 24 (+0.00%) | Shared Memory Configuration Size [Kbyte] | 8.19 (+0.00%) |
| Uses Green Context | 0 (+0.00%) | Stack Size | 1,024 (+0.00%) |
| # SMs [SM] | 24 (+0.00%) | # TPCs | 12 (+0.00%) |
| Enabled TPC IDs | all (all) | - | - |

## Occupancy

% Occupancy Graphs

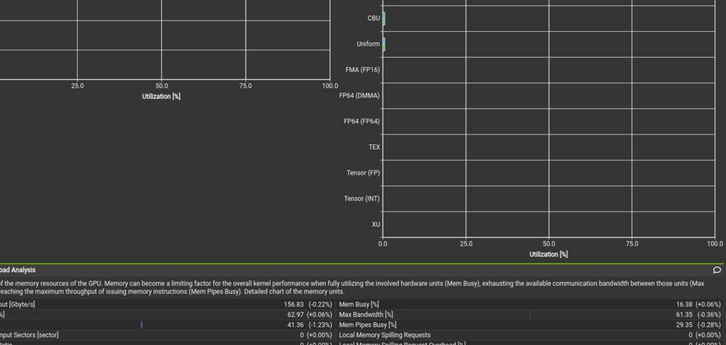
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

| | | | |
|---|---|---|---|
| Theoretical Occupancy [%] | 66.67 (+0.00%) | Block Limit Registers [block] | 2 (+0.00%) |
| Theoretical Active Warps per SM [warp] | 32 (+0.00%) | Block Limit Shared Mem [block] | 8 (+0.00%) |
| Achieved Occupancy [%] | 57.90 (+1.28%) | Block Limit Warps [block] | 1 (+0.00%) |
| Achieved Active Warps Per SM [warp] | 27.79 (+1.28%) | Block Limit SM [block] | 24 (+0.00%) |

**Theoretical Occupancy** The 8.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (66.7%) is limited by the number of warps within each block.
**Est. Local Speedup: 33.33%**

Key Performance Indicators







## GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

| | | | |
|---|---|---|---|
| Average SM Active Cycles [cycle] | 66,317.38 (+0.35%) | Average L1 Active Cycles [cycle] | 66,317.38 (+0.35%) |
| Average L2 Active Cycles [cycle] | 61,475.38 (+0.08%) | Average SMSP Active Cycles [cycle] | 64,255.01 (+1.06%) |
| Average DRAM Active Cycles [cycle] | 221,292 (-0.00%) | Total SM Elapsed Cycles [cycle] | 1,754,592 (+0.28%) |
| Total L1 Elapsed Cycles [cycle] | 1,754,592 (+0.28%) | Total L2 Elapsed Cycles [cycle] | 1,168,288 (+0.28%) |
| Total SMSP Elapsed Cycles [cycle] | 7,018,368 (+0.28%) | Total DRAM Elapsed Cycles [cycle] | 1,442,816 (+0.36%) |

### Workload Distribution

| | Average | Min | Max | Sum |
|---|---|---|---|---|
| SM Active Cycles | 66,317.38 (+0.35%) | 65,437 (+0.56%) | 66,818 (+0.14%) | 1,591,617 (+0.35%) |
| SMSP Active Cycles | 64,255.01 (+1.06%) | 62,959 (+2.46%) | 65,555 (+0.58%) | 6,168,481 (+1.06%) |
| L1 Active Cycles | 66,317.38 (+0.35%) | 65,437 (+0.56%) | 66,818 (+0.14%) | 1,591,617 (+0.35%) |
| L2 Active Cycles | 61,475.38 (+0.08%) | 61,036 (+1.00%) | 62,332 (+0.23%) | 983,606 (+0.08%) |
| DRAM Active Cycles | 221,292 (-0.00%) | 221,216 (+0.00%) | 221,328 (-0.01%) | 885,168 (-0.00%) |

## Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

| | | | |
|---|---|---|---|
| Branch Instructions [inst] | 36,816 (+0.00%) | Branch Efficiency [%] | 0 (+0.00%) |
| Branch Instructions Ratio [%] | 0.04 (+0.00%) | Avg. Divergent Branches [branches] | 0 (+0.00%) |

**Uncoalesced Global Accesses** This kernel has uncoalesced global accesses resulting in a total of 140944 excessive sectors (14% of the total 1023376 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.
**Est. Speedup: 11.60%**

Key Performance Indicators

### L2 Theoretical Sectors Global Excessive

| Location | Value | Value (%) |
|---|---|---|
| 0x72841f272550 in convolution3D_kernel | 17,618 | 13 |
| 0x72841f272540 in convolution3D_kernel | 17,618 | 13 |
| 0x72841f272530 in convolution3D_kernel | 17,618 | 13 |
| 0x72841f272520 in convolution3D_kernel | 17,618 | 13 |
| 0x72841f272510 in convolution3D_kernel | 17,618 | 13 |

### Warp Stall Sampling (All Samples)

| Location | Value | Value (%) |
|---|---|---|
| 0x72841f272570 in convolution3D_ke... | 350 | 20 |
| 0x72841f272580 in convolution3D_ke... | 313 | 20 |
| 0x72841f2725e0 in convolution3D_ke... | 126 | 8 |
| 0x72841f272560 in convolution3D_ke... | 124 | 8 |
| 0x72841f2725d0 in convolution3D_ke... | 86 | 6 |

### Most Instructions Executed

| Location | Value | Value (%) |
|---|---|---|
| 0x72841f272430 in convolution3D_ke... | 18,432 | 4 |
| 0x72841f272410 in convolution3D_ke... | 18,432 | 4 |
| 0x72841f272400 in convolution3D_ke... | 18,432 | 4 |
| 0x72841f272310 in convolution3D_ker... | 18,432 | 4 |
| 0x72841f272340 in convolution3D_ke... | 18,432 | 4 |

Still missing data? Collect the full metric set, enable individual sections or learn how to select individual metrics for collection.